AFE4950 – IEC Testing Report

Medical Business Unit, Texas Instruments

OVERVIEW

This document describes the results of testing the AFE4950 as per ANSI/AAMI/IEC 60601-2-47:2012/(R)2016, which specifies the requirements for the performance of ambulatory electrocardiographic (ECG) systems. The tests are described in Section ***201.12.4 Protection against hazardous output*** of ANSI/AAMI/IEC 60601-2-47:2012/(R)2016 specification.

Document History

Version	Date	Notes
1.0	30 March 2020	First release
1.1	10th April 2020	Added Gain Stability Test result

Contents

1 Introduction 2
2 Test Results 3
2.1 Incorrect output (201.12.4.4) 3
2.1.1 Linearity and dynamic range (201.12.4.4.101) 3
2.1.2 Input impedance (201.12.4.4.102) 4
2.1.3 Common Mode Rejection (201.12.4.4.103) 5
2.1.4 Gain Accuracy (201.12.4.4.104) 6
2.1.5 Gain Stability (201.12.4.4.105) 7
2.1.6 System Noise (201.12.4.4.106) 8
2.1.7 Multichannel crosstalk (201.12.4.4.107) 8
2.1.8 Frequency Response (201.12.4.4.108) 9
2.1.9 Function in the presence of pacemaker pulses (201.12.4.4.109) 12
2.1.10 Timing Accuracy (201.12.4.4.110) 13
2.1.11 Gain settings and switching (201.12.4.4.111) 13
2.1.12 Temporal Alignment (201.12.4.4.112) 13

1 Introduction

The full system-level IEC test of a typical ECG recording system using the AFE4950 is depicted in Figure 1. The AFE-level tests results described in this document however involve only a subset of the full system-level test, specifically only the blocks shown in grey.

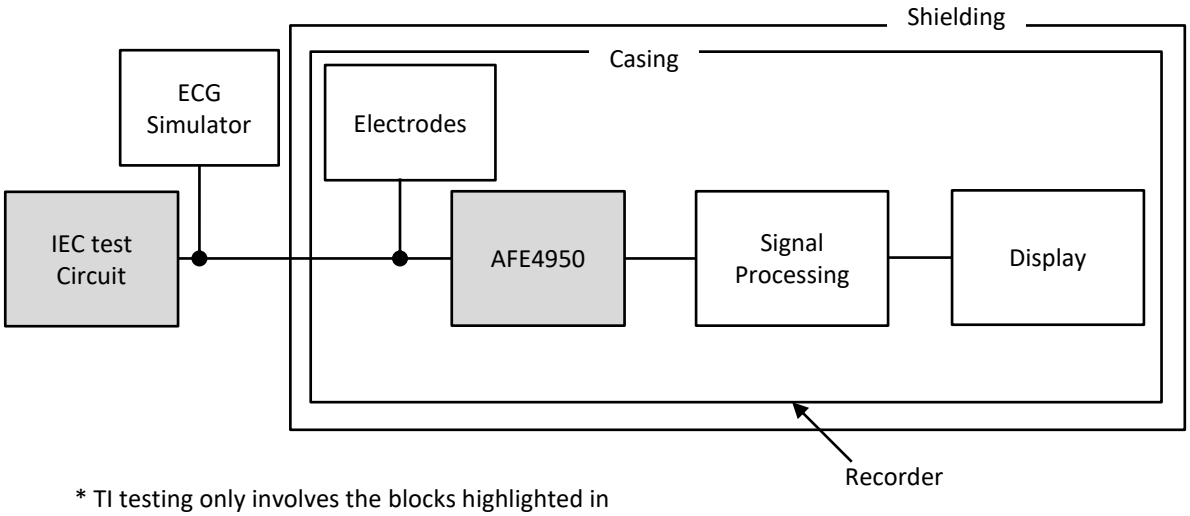


Figure 1. Block diagram of IEC testing of an ECG recorder system

2 Test Results

2.1 Incorrect output (201.12.4.4)

2.1.1 Linearity and dynamic range (201.12.4.4.101)

This test measures the performance and reproducibility of the ECG signal chain in the presence of ±300mV DC differential offset and varying amplitudes (0.5mV, 1mV, 2mV and 6mV). The measurement output should not vary by more than 10% or 50µV whichever is greater.

2.1.1.1 Test and Result:

A 10.4Hz triangular waveform was input to the AFE and the output voltage amplitude was deduced from the output code. The output voltage was divided by the nominal value of the set INA gain to get the measured amplitude.

Table 1. Measured amplitude in linearity and dynamic range test, INA gain = 21

DC Offset (V)	Input Amplitude (mV)	Measured Amplitude (mV)	Absolute Error (µV)
0	0.5	0.4996	0.35
0	1	0.9983	1.72
0	2	1.9963	3.71
0	6	5.9934	6.63
0.3	0.5	0.4997	0.26
0.3	1	0.9988	1.21
0.3	2	1.9968	3.15
0.3	6	5.9948	5.19

2.1.1.2 Summary:

The measured amplitude is well within specification of 10% or 50µV for all conditions.

2.1.2 Input impedance (201.12.4.4.102)

This test measures the input impedance of the AFE at 10Hz in the presence of ±300mV DC differential offset. 10Hz signal is measured with and without a series impedance of 620kΩ||4.7nF and output amplitude variation should be less than 6%, which ensures that the input impedance is greater than 10MΩ.

2.1.2.1 Test and Result:

In the wearable applications the electrode impedance will be much higher than 620kΩ||4.7nF. So AFE4950 was tested with a more stringent electrode impedance of 10MΩ and a signal frequency of 60Hz. A 60Hz sinusoidal input signal of 5mV p-v amplitude is measured with and without a series impedance of 10MΩ. Measurement results of the amplitude error are listed in Table 2.

Table 2: Input impedance measurement results, with an input of 5mV, 60Hz sinusoidal signal.

DC Offset (V)	Series Impedance	Amplitude Error (%)
0	0Ω	0
0	10MΩ	-0.54
0.3	0Ω	0.11
0.3	10MΩ	-0.61

2.1.2.2 Summary:

Maximum amplitude error is <1% even with a series impedance of 10MΩ, well within the specification of 6%. Impedance of the device at 60Hz is 3.7GΩ with DC offset = 0 and 2.8GHz with DC offset = 300mV. At 10Hz the input impedance is > 10GΩ

2.1.3 Common Mode Rejection (201.12.4.4.103)

This test measures the common mode rejection (CMRR) performance of the AFE for 50Hz and 100Hz interference. CMR for 50Hz should be at least 60 dB and for 100Hz it should be at least 45 dB.

2.1.3.1 Test and results:

50Hz CMR is measured by applying a 4V p-v 50Hz sinusoidal signal (8V p-v signal with a coupling of 100pF from the source and attenuation of 0.5). 100Hz CMRR is measured by applying a 0.711V p-v 100Hz sinusoidal signal (1.422V signal with a coupling of 100pF from source and attenuation of 0.5). CMRR Measurement results for 60Hz and 120Hz coupling are listed in Table 3.

Table 3: 3-electrode Common Mode Rejection for 50Hz and 100Hz

DC Offset (V)	Impedance	CMRR (dB) 60Hz	CMRR (dB) 120Hz
0	51kΩ||47nF on both inputs	143.4	140.4
0	51kΩ||47nF asymmetry between electrodes	141.8	138.7
0.3	51kΩ||47nF on both inputs	140.7	130.5
0.3	51kΩ||47nF asymmetry between electrodes	138.4	120.1

Table 4: 2-electrode Common Mode Rejection for 50Hz and 100Hz

DC Offset (V)	Impedance	CMRR (dB) 60Hz	CMRR (dB) 120Hz
0	51kΩ||47nF on both inputs	109.5	109.7
0	51kΩ||47nF asymmetry between electrodes	81.9	82.1
0.3	51kΩ||47nF on both inputs	99.6	95.7
0.3	51kΩ||47nF asymmetry between electrodes	81.8	81.9

2.1.3.2 Summary:

For 3-electrode configuration, common mode rejection at 60Hz and 120Hz it is greater than 120 dB and for 2-electrode configuration, common mode rejection is greater than 81dB.

2.1.4 Gain Accuracy (201.12.4.4.104)

This test measures the gain accuracy of the device. Gain error should be within ±10% for all possible gain settings.

2.1.4.1 Test and results:

Gain is measured by applying 5Hz, 2mV p-v sinusoidal signal. Measurement results of gain accuracy are listed in Table 5.

Table 5: Gain Accuracy

Gain	Measured Gain	Gain Accuracy (%)
11	10.980	-0.180
21	20.961	-0.185

2.1.4.2 Summary:

The gain accuracy meets the specification for all gain settings.

2.1.5 Gain Stability (201.12.4.4.105)

This test measures the AFE gain stability over 24 hours. The gain change should be less than 3% over 24 hours.

2.1.5.1 Test and Results:

Gain is measured with a 5Hz, 2mV p-v sinusoidal signal. Gain is monitored at 1m, 2m, 5m, 10m, 20m, 30m, 45m, 60m and once every hour for the following 24hours. Measurement results of gain stability are shown in

.

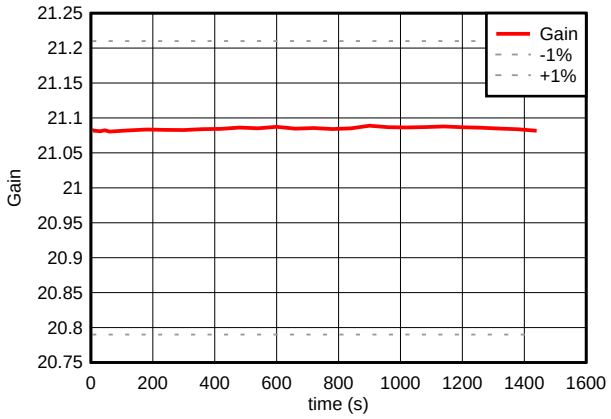


Figure 2. Gain stability measurement over 24 hours

2.1.5.2 Summary:

AFE gain stability is well within the spec.

2.1.6 System Noise (201.12.4.4.106)

Input referred noise should not exceed 50 µV p-v over any 10s period when inputs are connected through 51kΩ||47nF in series with each electrode.

2.1.6.1 Test and Result:

Noise measurement results are shown in Figure 3. Peak to peak noise is <6uVp-p. These measurements are taken with a PRF of 1000Hz and 150Hz filtering is applied externally.

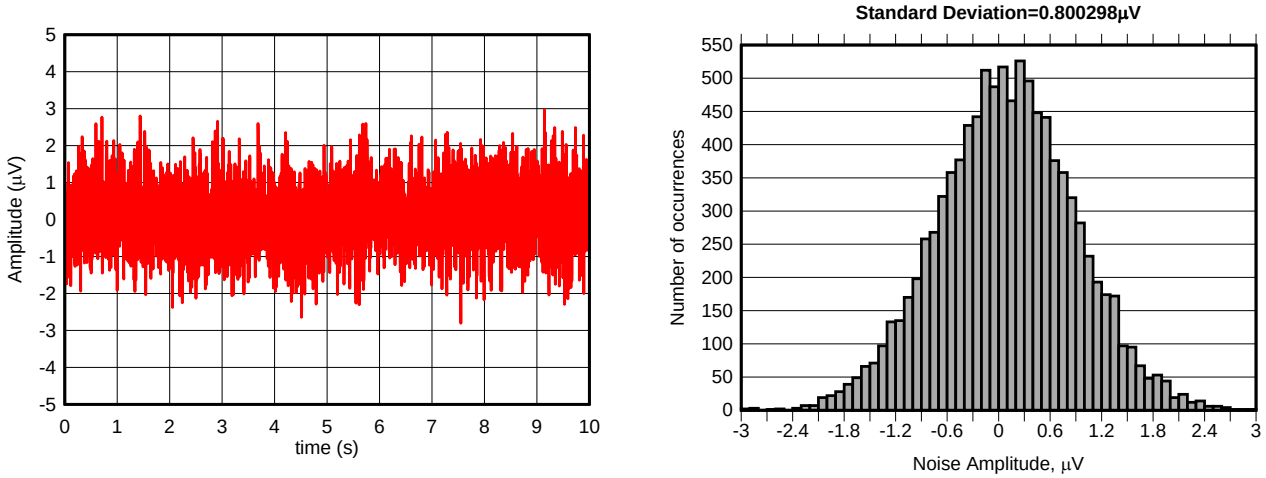


Figure 3. System noise over 0.5 to 150Hz, 51kΩ||47nF in series with each electrode, ECG sample rate = 1kHz.

2.1.6.2 Summary:

Noise of the AFE is well within the spec.

2.1.7 Multichannel crosstalk (201.12.4.4.107)

N/A

2.1.8 Frequency Response (201.12.4.4.108)

Frequency response test specifies the response for three different types of inputs.

2.1.8.1 Rectangular Pulse Input

The characteristics of the response to a 3 mV, 100 ms rectangular pulse are specified. The baseline amplitude displacement after the pulse should be less than 0.1 mV referred to the baseline before the pulse. The slope outside the pulse should be less than 0.3 mV/s and the leading edge overshoot should be less than 10%.

2.1.8.1.1 Test and Result:

Response to the rectangular pulse input is shown in Figure 6. The AFE has an integrated high pass filter. The corner of this HPF set by an external capacitor. For this measurement, the HPF corner frequency is set as 0.5Hz. Due to the 0.5Hz high pass filter, the rectangular pulse will have a 26.9% droop in 100ms ($1 - e^{(-0.1s/\tau_{HPF})}$ = 0.269). Because of this droop, the undershoot at the rectangular pulse falling edge is also high (0.85mV). If the AFE is configured to have 0.05Hz high pass filter, then the droop will be 3.1% and undershoot will be < 0.1mV.

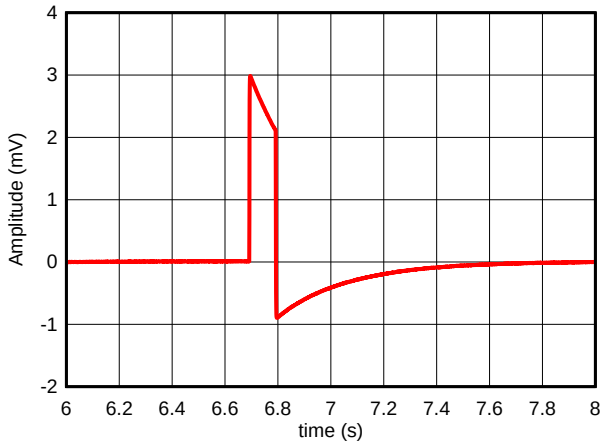


Figure 4. Response to a rectangular pulse, with f_{HPF} = 0.5Hz

2.1.8.1.2 Summary:

For an AFE with high pass filter, to meet the rectangular pulse response, the filter corner frequency has to be lower than 0.05Hz. HPF frequency of AFE4950 is set by an external capacitor connected between ECG_CAPP and ECG_CAPM pins, which allow setting a 0.05Hz high pass filter corner frequency.

2.1.8.2 Sinusoidal Frequency Response

Amplitude response for the frequency range of 0.67Hz to 40Hz shall be within 70 % and 140 % of the response at 5Hz.

2.1.8.2.1 Test and Result:

Frequency response is measured by applying a 2mV p-v sinusoidal signal with different frequencies in the range of 0.5Hz to 500Hz. The sinusoidal frequency response is shown in Figure 5.

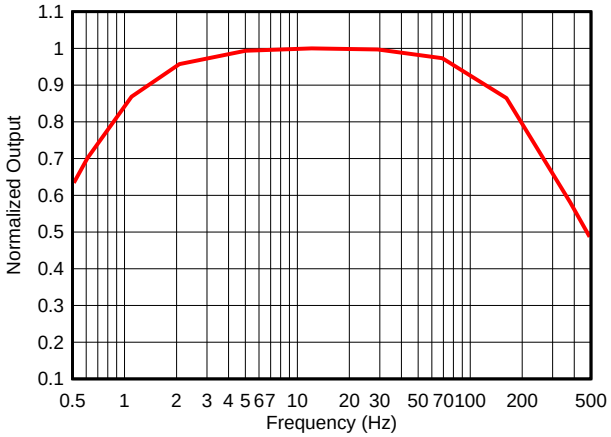


Figure 5. Frequency response to sinusoidal input.

2.1.8.2.2 Summary:

The sinusoidal frequency response is within spec.

2.1.8.3 Triangular Input

Response to 1.5mV, 40ms triangular pulse train which simulates a series of narrow R-waves, should be within 70% to 110% of the maximum amplitude in a train of 1.5mV, 200ms triangular pulses.

2.1.8.3.1 Test and Result:

The response to the triangular pulse train input is shown in Figure 6. Peak to peak amplitude Error of 40ms triangular pulse train with respect to 200ms pulse train is 0.9%. Positive peak of 40ms triangular pulse has an error of -4.46% and negative peak has an error of -3.12% with respect to 200ms pulse.

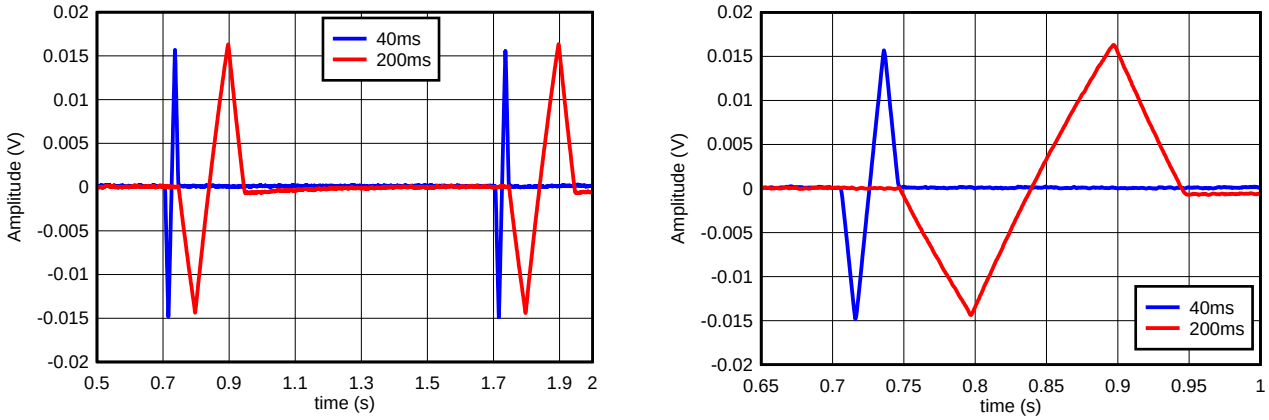


Figure 6. Response to a Triangular pulse

2.1.8.3.2 Summary:

Triangular input response for different pulse widths meets the spec.

2.1.9 Function in the presence of pacemaker pulses (201.12.4.4.109)

This test measures the performance of the ECG device in the presence of pacemaker signals.

2.1.9.1 Test and Result:

The ability of recording ECG signals in the presence of implanted pacemaker pulses is checked by applying a 2mV, 10 Hz sinusoidal signal and adding a sharp pulse train (200 mV pulse of duration 1 ms and rise time <100 µs) repeating at 100 pulses/min. The height of the second peak of the sine wave after the pulse should not differ by more than 0.2 mV from the height of the sine wave peak immediately preceding the pulse. Figure 7 shows the response in the presence of the pacemaker signal. The height of the second sine wave peak has an error of 0.22mV.

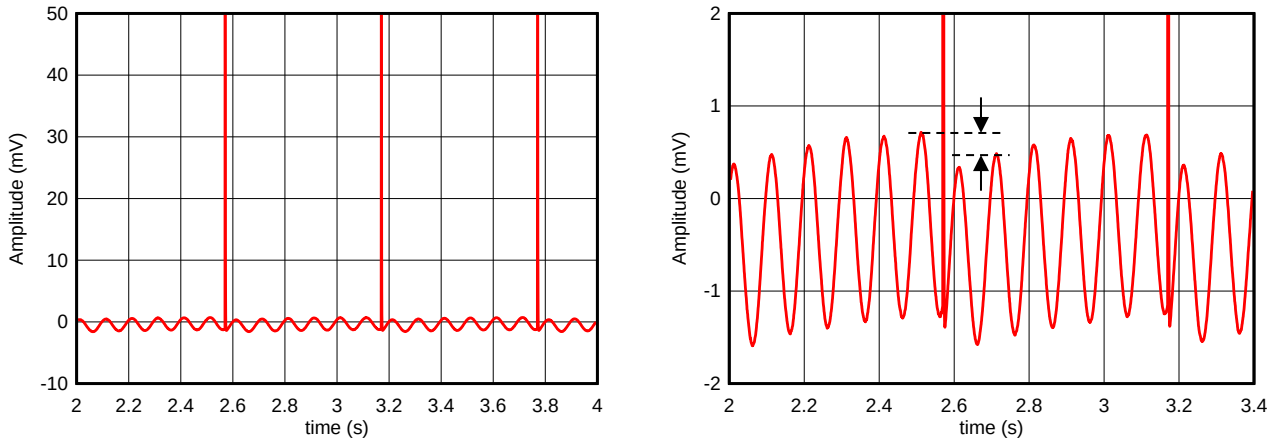


Figure 7. Response in the presence of pacemaker pulse

2.1.9.2 Summary:

This spec is also affected by the high pass filter corner frequency. Using a lower high pass filter corner frequency helps to meet to the spec.

2.1.10 Timing Accuracy (201.12.4.4.110)

The timing accuracy measures the stability of the output sample rate over 24hours duration. At 1h, 8h and 23h the time stamp of the output sample should be with in 30s of an ideal system.

The accuracy of this spec depends on the accuracy of the clock. AFE should be used with accurate external clock input of 32.768 kHz to meet this spec.

2.1.11 Gain settings and switching (201.12.4.4.111)

N/A

2.1.12 Temporal Alignment (201.12.4.4.112)

N/A